Suggested Teaching Guidelines for

Database Technologies

PGCP-AC, August 2026

Duration: 90 Hours: 30 Theory (T) + 30 Lab (L) + 30 Self Learning (SL)

Objective: To equip students with the skills to design, query, program, and manage relational and NoSQL databases using MySQL and MongoDB.

Prerequisites: Basic Knowledge of Programming Concepts.

Evaluation method: Theory exam – 40%, Lab: exam – 40%, Internal Assessment – 20%

List of Course Material

Text book:

1. The Language of MySQL, 3rd edition, Larry Rockoff / Pearson

Reference book:

1. Murach's MySQL, 4th Edition, Joel Murach / Murach Books
2. MongoDB: The Definitive Guide, 3rd Edition, Shannon Bradshaw, Eoin Brazil, Kristina Chodorow / O'Reilly

Session 1: Introduction to DBMS and SQL (2T + 4L + 2SL)

- Basic Database Terminologies
- Types of DBMS: Relational, Object Relational and NoSQL Databases
- Introduction to MySQL, MySQL Clients (Monitor, Shell, Workbench)
- Data Models (Conceptual, Logical, Physical)
- Database Design, Entity-Relationship Diagram
- Categories of SQL Commands: DDL, DML, DCL, DTL/TCL
- Data Definition Language (DDL)
  - CREATE Statement
  - ALTER Statement
  - DROP Statement
  - TRUNCATE Statement
  - Difference between DELETE and TRUNCATE
  - RENAME Statement

Lab:

- Using MySQL Monitor, Shell, and Workbench
- Performing basic CREATE, ALTER, DROP Commands

Session 2: Normalization (2T + 2SL)

- Data Redundancy and Data Anomalies
- Functional Dependency
- Introduction to Normalization

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- Need for Normalization
- Normal Forms (1st NF, 2nd NF, 3rd NF, BCNF) with examples
- Introduction to 4th and 5th NF

Session 3: DML Commands (2T + 2L + 2SL)

- Introduction to DML Commands
- Difference between DDL and DML Commands
- Inserting Data into Tables
- Updating Existing Data
- Deleting Data from Tables
- Difference between DELETE and TRUNCATE
- Data Integrity Considerations while Modifying Data

Lab:

- Create tables and insert records using INSERT statements.
- Insert multiple records into a table.
- Update specific records using UPDATE and WHERE clauses.
- Delete selected records using DELETE statement.
- Delete all records from a table using DELETE.
- Use TRUNCATE to remove all rows from a table.
- Compare the behavior of DELETE and TRUNCATE commands.

Session 4: Data Types, Constraints and Aggregate Functions (2T + 2L + 2SL)

- MySQL Data Types, Database Constraints (Primary Key, Unique, Not Null, Foreign Key, Default, Check*)
- Aggregate Functions, Grouping Things Together (Group By, Having)
- LIKE Operator, DISTINCT, Sorting (Order by clause)
- BETWEEN… AND Operators, Comparing Nulls (IS NULL/IS Not NULL), IN/NOT IN

Lab:

- Defining Data Types for Columns
- Creating, Altering, Dropping Constraints
- Aggregate Functions: SUM(), AVG(), COUNT(), MAX(), MIN(), COUNT(), Group By, Having Clause
- Using Like, Distinct, Order By, Between...And
- Comparing Nulls, Using IN/Not-In

Session 5: Relational Algebra and Joins (2T + 2L + 4SL)

- Relational Algebra Operations (Selection, Projection, Union, Intersect*, Minus*, Cross/Cartesian)
- Joins (Eqvi, Inner, Outer, Natural, Cross), SQL Standard Syntax for Joins

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- Copying table structure/data, Sequences (AUTO_INCREMENT)

Lab:

- Union/Union ALL
  Queries on Various type of Joins using OLD and SQL Standard Syntax
- Copying table structure, Copying data from one table to another
- Using AUTO_INCREMENT

Session 6: Window Functions and CTEs (2T + 2L + 2SL)

- Introduction to Window Functions: concept and use over GROUP BY
- ROW_NUMBER (), RANK (), DENSE_RANK (), NTILE ()
- Aggregate window functions: SUM OVER, AVG OVER, running totals
- PARTITION BY and ORDER BY within OVER clause
- CTEs: WITH clause syntax, simple and chained CTEs
- Recursive CTEs for hierarchical/tree data

Lab:

- Write queries using ROW_NUMBER, RANK, DENSE_RANK to rank employees
- Running totals and moving averages using SUM OVER with PARTITION BY
- Rewrite complex subqueries as CTEs for readability; recursive CTE for org hierarchy

Session 7: Subqueries, TCL, DCL, Views & Indexes (3T + 3L + 2SL)

- Subquery, Correlated Subquery, EXISTS/NOT EXISTS
- TCL Commands (Commit/Rollback/Savepoint), DCL Commands (GRANT/REVOKE/GRANT OPTION)
- Views, Types of Views, Simple and Complex Views
- Indexes, Benefit of Indexes, Type of Indexes, Temporary Tables
- ACID Properties, Concept of Database Instance and Schema
- MySQL Storage Engines (InnoDB, MyISAM and others)

Lab:

- Subqueries, Correlated Queries
- Using Exists/Not-Exists
- Using Commit/Rollback/Savepoint
- Granting/revoking privileges on database objects
- Creating Views, Querying using Views
- Creating Indexes
- Creating Temporary Tables
- Indexes, Temporary Tables

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Session 8: Stored Procedures (2T + 2L)

- Introduction to MySQL Programming, Use of MySQL Programs,
- Introduction to Stored Procedures, Benefits of Stored Procedures
- Procedure Parameters (IN, OUT and INOUT).

Lab:

- Creating procedure without parameters
- Creating Procedure with (IN/OUT/INOUT) Parameters

Session 9: (2T + 2L+ 2SL)

- Flow Control Statements (LOOP, WHILE and REPEAT)
- Using above statements in Stored Procedures/Functions
- Conditional Statements (IF, IF-ELSE-THEN, SWITCH CASE)
- Example of each type of statement
- Loop constructs (ITERATE, LEAVE)

Lab:

- Use of flow control statement in Stored Procedure
- Use of conditional statements in Stored Procedure

Session 10: Functions (2T + 2L)

- Functions with and without parameters
- MySQL Built-in functions (string, numeric, date etc.)

Lab:

- Creating Function and returning value from it
- Use of built-in functions in queries

Session 11: Cursors (2T + 2L)

- Introduction to Cursors
- Cursor Lifecycle
  - DECLARE
  - OPEN
  - FETCH
  - CLOSE
- Types of Cursors
  - Asensitive Cursor
  - Insensitive Cursor
  - Read-Only Cursor
  - Non-scrollable Cursor

- Real-World Cursor Examples

Lab:
- Writing procedures with Declare, fetch and close cursor
- Create and execute a simple cursor-based stored procedure.
- Declare, open, fetch, and close a cursor.
- Process records one row at a time using a cursor.
- Use cursors with loops and conditional statements.
- Implement a cursor with exception/handler support.

Session 12: Triggers (1T + 1L + 2SL)

- Introduction to Triggers
- Types of Triggers: BEFORE, AFTER, Row-level
- Trigger variables
- Real-World Trigger Examples

Lab:
- Create Before Triggers and After Triggers

Session 13: Error Handling and Exception Management (2T + 2L + 2SL)

- Error Handling and Exceptions
- Types of Handler Actions
- Writing Handlers
- Defining and handling exceptions in Stored Procedures and Functions

Lab:
- Exception handling in Stored Procedure
- Exception handling with various handler actions

Session 14: Introduction to NoSQL (2T + 2L + 2SL)

- Features of NoSQL Database
- Structured vs. Semi-structured and Unstructured Data
- Difference between RDBMS and NoSQL databases
- CAP Theorem, BASE Model
- Categories of NoSQL Databases
  - Key-Value Store
  - Document Store
  - Column-Oriented
  - Graph
- Introduction to MongoDB, Features of MongoDB
- MongoDB command interface and MongoDB compass

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- MongoDB Documents & Collections
- RDBMS & MongoDB analogies
  - relations/tables => collections; tuples/records => documents
- JSON and BSON documents

Session 15: Getting Started with MongoDB (2T + 2L + 6SL)

- CRUD operations: insertOne / insertMany, find / findOne, updateOne / updateMany / replaceOne, deleteOne / deleteMany
- UPSERT: update with upsert: true option
- Query operators: comparison ($eq, $gt, $lt, $in, $nin), logical ($and, $or, $not, $nor)
- Update operators: $set, $unset, $inc, $push, $pull, $addToSet
- Sorting with sort (); pagination with skip () and limit ()
- Indexes in MongoDB: createIndex(), compound indexes, unique indexes

Lab:

- Creating database, connecting to a database, Creating Collections
- Performing CRUD operations: insert, find with filters, update, delete
- Sorting, pagination (skip/limit), UPSERT